Tech Interview Prep:

Virtualization vs. Containers and Infrastructure as Code

Section 1: Virtual Machines vs. Containers

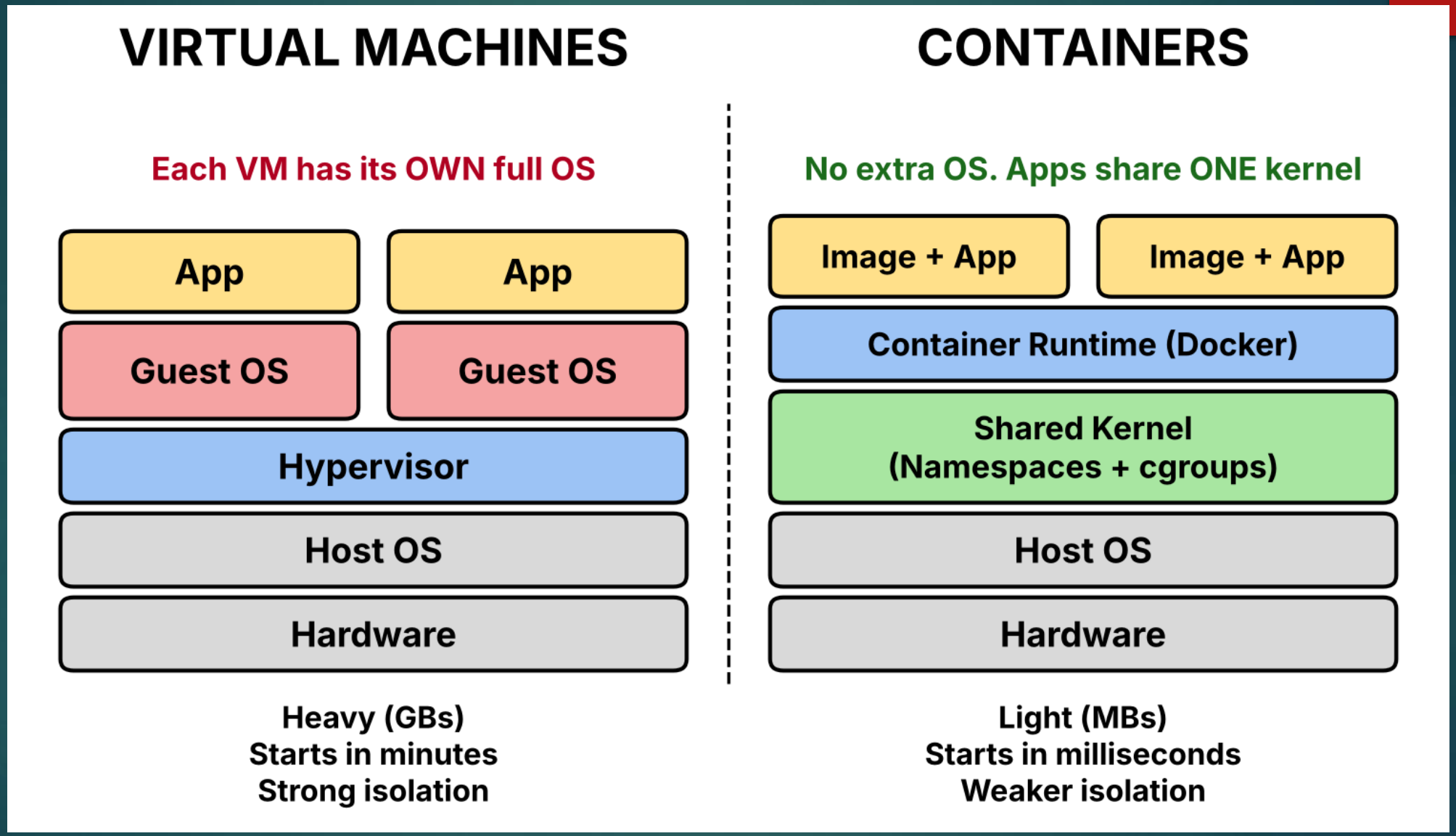
Section 2: Infrastructure as Code (IaC)

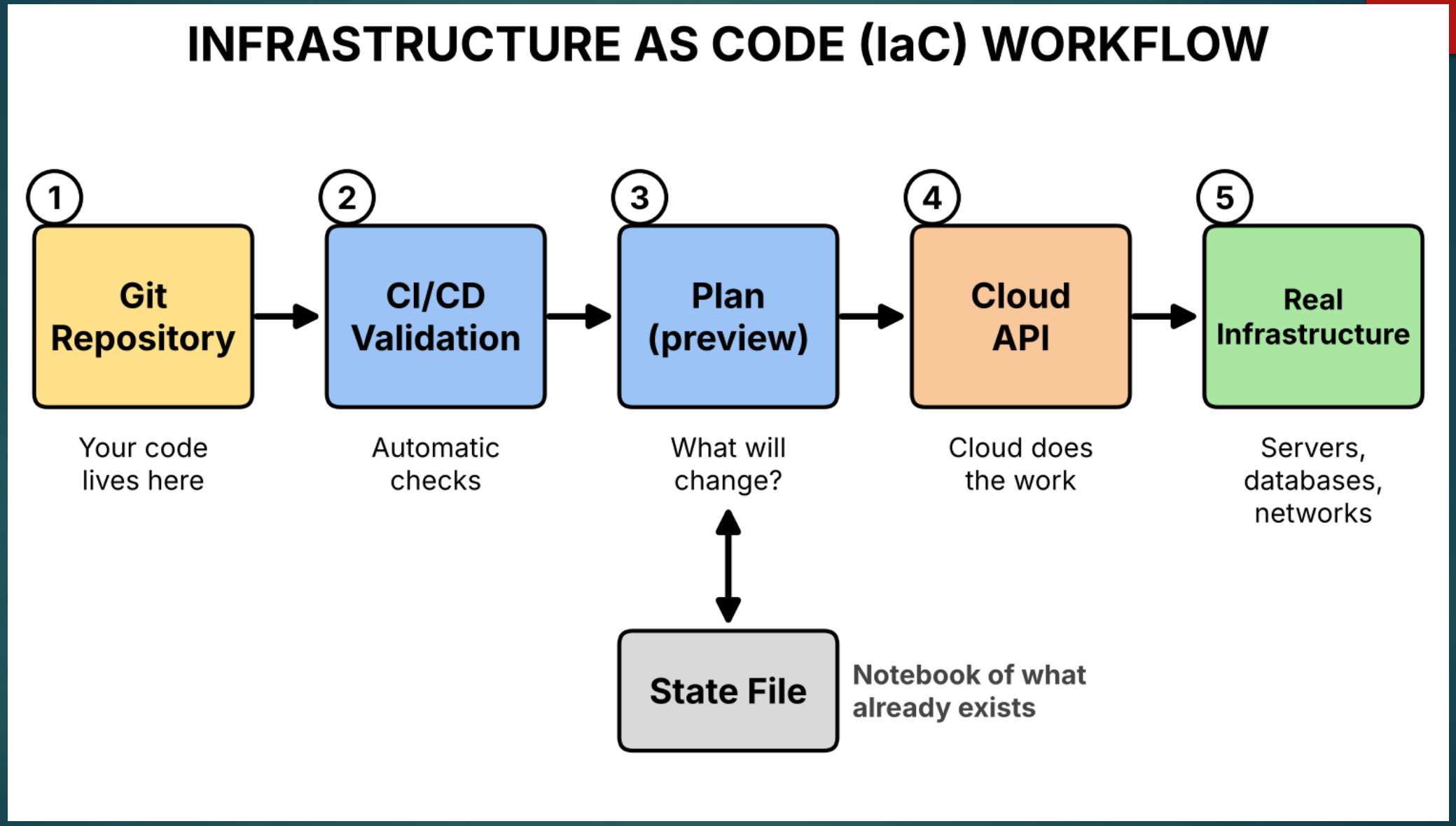
Section 3: Synthesis & When to Use Which

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CLASS: AD371 (APPLICATION ARCHITECTURE)

Section 1: Virtual Machines vs. Containers




INFRASTRUCTURE AS CODE (IaC) WORKFLOW
1
Git Repository
Your code lives here
2
CI/CD Validation
Automatic checks
3
Plan (preview)
What will change?
4
Cloud API
Cloud does the work
5
Real Infrastructure
Servers, databases, networks
State File
Notebook of what already exists

Section 3: Synthesis & When to Use Which

- Containers vs. Virtual Machines (fast, light, easy to scale vs. strong isolation)
- Infrastructure as Code → instead of building servers by hand, write down what we want in a file, and a tool builds it for us.